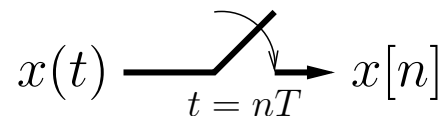


Sampling and Reconstruction:

- Until now, we have considered continuous-time signals & systems *separately* from discrete-time signals & systems.
- We will now connect them through *uniform sampling*:

$$x[n] = x(nT) \text{ for } n \in \mathbb{Z},$$

where T (in seconds/sample) is the *sampling interval* and so $1/T$ is the *sampling rate*.



- Key question:
 - Do we always lose information in going from $\{x(t)\}_{\forall t}$ to $\{x[n]\}_{n=-\infty}^{\infty}$?
 - Under what conditions can we reconstruct $\{x(t)\}_{\forall t}$ from $\{x[n]\}_{\forall n}$?

Aliasing — intuitions:

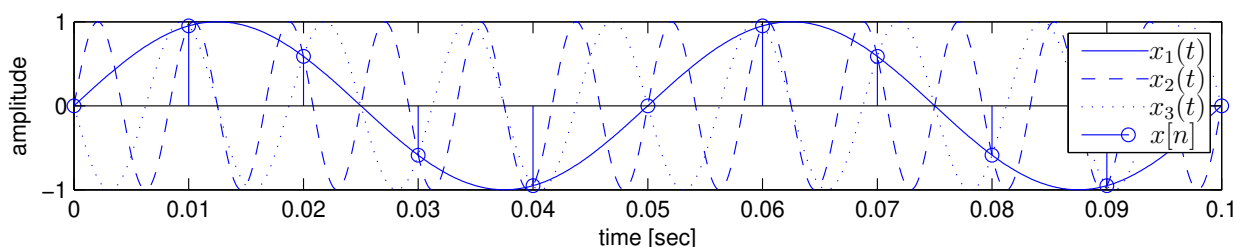
- Consider sampling the 3 sinusoids

$$x_1(t) = \sin(2\pi 20t)$$

$$x_2(t) = \sin(2\pi 120t)$$

$$x_3(t) = \sin(2\pi(-80)t)$$

at a rate of $1/T = 100$ samples/sec to give $x[n]$:



They all yield the same $x[n]$!

- More generally, consider sampling a complex sinusoid:

$$x(t) = e^{j\Omega_k t} \xrightarrow{t=nT} x[n] = e^{j\Omega_k T n}.$$

Which different Ω_k yield identical sequences $\{x[n]\}$?

Consider

$$\Omega_k \triangleq \Omega_0 + \frac{2\pi k}{T} \text{ for any integer } k.$$

Then

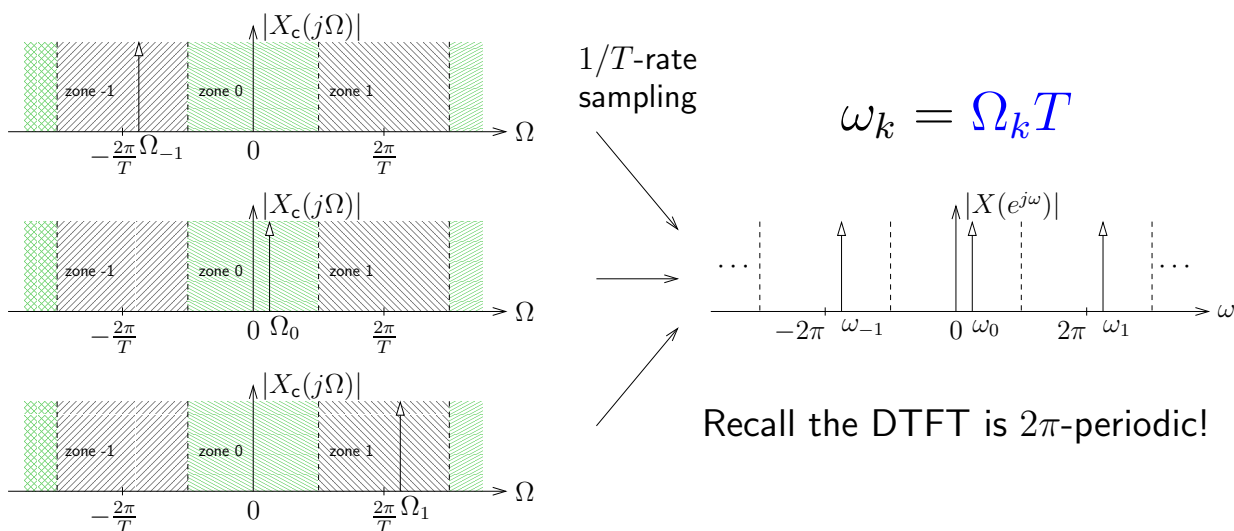
$$e^{j\Omega_k T n} = e^{j(\Omega_0 T n + 2\pi k n)} = e^{j\Omega_0 T n} = e^{j\omega_0 n},$$

where $\boxed{\omega_0 = \Omega_0 T}$, for any value of k .

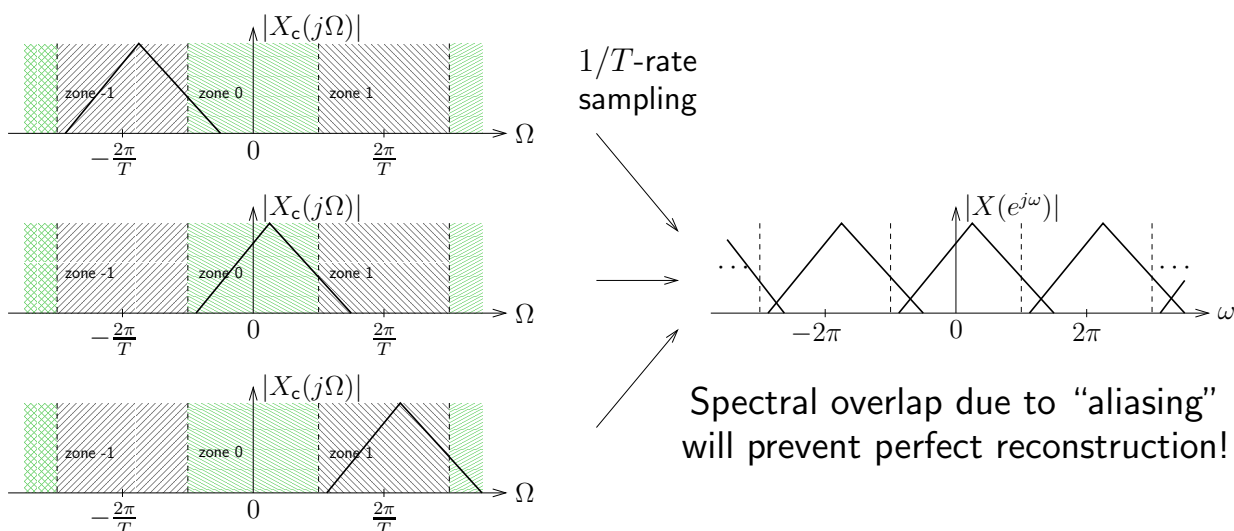
Aliasing — Nyquist zones:

- So far, we have seen that a sampled sinusoid $x[n]$ can be associated with *many* continuous-time sinusoids $x(t)$.
- Can visualize this for $x(t) = e^{j\Omega_k t}$ via “Nyquist zones”:

Defn: The k^{th} Nyquist zone is $\Omega \in (\frac{2\pi k - \pi}{T}, \frac{2\pi k + \pi}{T}]$



- Due to the linearity of sampling, the same phenomenon applies to generic signals:



- Nyquist zone locations determined by sampling rate $\frac{1}{T}$!

Nyquist sampling:

- As we have seen, sampling can introduce ambiguity:

Can't distinguish one Nyquist zone from another!

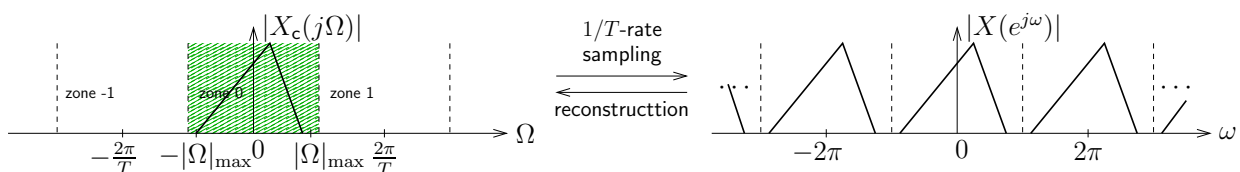
- How can we avoid this ambiguity?

Force the CT signal to live in a single Nyquist zone via careful choice of sampling frequency $1/T$.

There are two basic ways to do this:

1. Nyquist sampling:

- Ensure that all frequency components of $x(t)$ are in the 0^{th} Nyquist zone. Equivalently...
- Sample more than twice as fast as the largest absolute signal frequency: $\frac{1}{T} > \frac{|\Omega|_{\max}}{\pi} = 2|f|_{\max}$

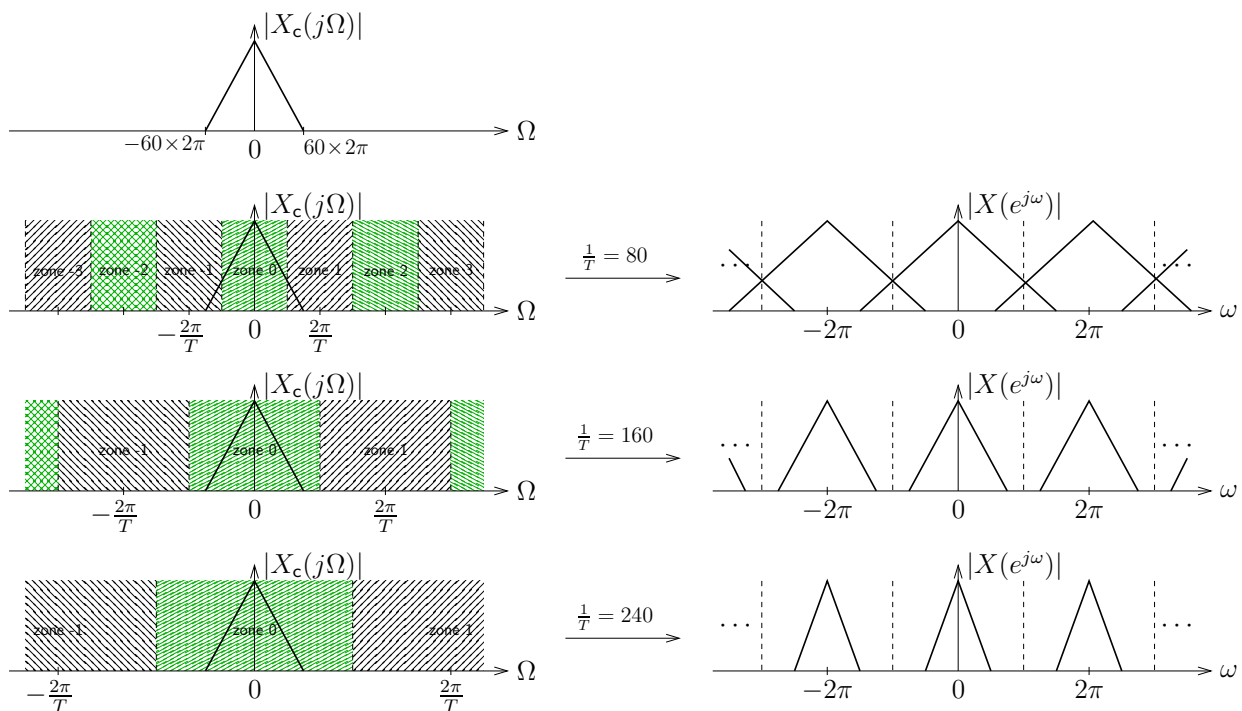


2. Bandpass sampling:

- Ensure that all frequency components of $x(t)$ are contained within some Nyquist zone k , for $k \neq 0$.
 - Note: works only for bandpass $x(t)$.
 - But how does one choose $\frac{1}{T}$ and k in this case?
- Reconstruction then boils down to optimal interpolation. Stay tuned for details...

Nyquist sampling – example:

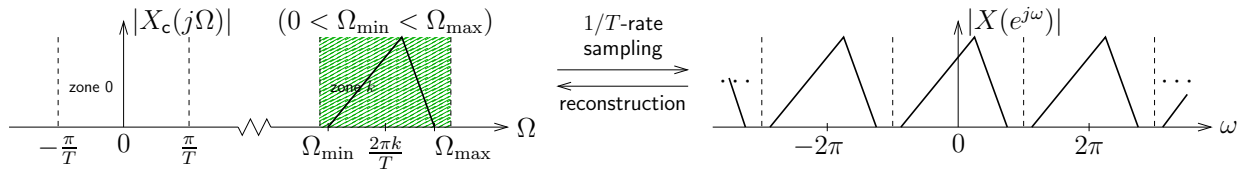
- Suppose that $X_c(j\Omega)$ has the CTFT magnitude sketched below, where $|f|_{\max} = 60$ Hz.
- Nyquist sampling specifies that $\frac{1}{T} > 2 \times 60 = 120$ Hz.
- The figures below illustrate the Nyquist zones and the resulting DTFT for several choices of sampling rate $\frac{1}{T}$.



- Note the presence of aliasing in the DTFT when $\frac{1}{T}$ is less than the Nyquist sampling rate (i.e., $2|f|_{\max}$ Hz).

Bandpass sampling – complex-valued signals:

- Consider a complex-valued bandpass signal of the form



- For practical reasons, we want to find the *smallest* possible sampling freq $\frac{1}{T}$. Then how do we choose k ?
- First, note that fitting into the k^{th} Nyquist zone requires

$$\left\{ \begin{array}{l} \frac{2\pi k - \pi}{T} < \Omega_{\min} \\ \Omega_{\max} \leq \frac{2\pi k + \pi}{T} \end{array} \right\} \iff \frac{k-0.5}{f_{\min}} < T \leq \frac{k+0.5}{f_{\max}}.$$

- Also, note that T increases ($\frac{1}{T}$ decreases) as k increases. Thus, we want to find the largest integer k that satisfies

$$\frac{k-0.5}{f_{\min}} < \frac{k+0.5}{f_{\max}},$$

and then use this “ $k_{\max}$ ” to set the sampling interval as

$$T = \frac{k_{\max} + 0.5}{f_{\max}}.$$

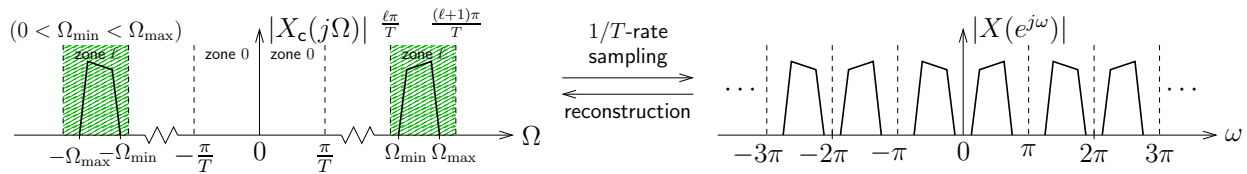
- After a bit of algebra, one can find

$$\frac{1}{T} \Big|_{\min} = \frac{f_{\max}}{\left\lceil \frac{1}{2} \frac{f_{\max} + f_{\min}}{f_{\max} - f_{\min}} \right\rceil - \frac{1}{2}}$$

where $\lceil \cdot \rceil$ denotes the “ceiling” operation, i.e., rounding up to the nearest integer.

Bandpass sampling – real-valued signals:

- Consider a real-valued bandpass signal of the form



Want “half-spectrum” to fit into a “half-Nyquist” zone. These zones are width- $\frac{\pi}{T}$ and mirrored across $\Omega = 0$.

- Again, want to find smallest possible sampling freq $\frac{1}{T}$.
- Fitting into the ℓ^{th} half-Nyquist zone implies

$$\left\{ \begin{array}{l} \frac{\ell\pi}{T} < \Omega_{\min} \\ \Omega_{\max} \leq \frac{(\ell+1)\pi}{T} \end{array} \right\} \Leftrightarrow \frac{\ell}{2f_{\min}} < T \leq \frac{\ell+1}{2f_{\max}}$$

- Note that T increases ($\frac{1}{T}$ decreases) as ℓ increases. Thus, we want to find the largest integer ℓ that satisfies

$$\frac{\ell}{2f_{\min}} < \frac{\ell+1}{2f_{\max}},$$

and then use this “ $\ell_{\max}$ ” to set the sampling interval as

$$T = \frac{\ell_{\max}+1}{2f_{\max}}.$$

- After a bit of algebra, one can find

$$\frac{1}{T} \Big|_{\min} = \left\lceil \frac{2f_{\max}}{f_{\max} - f_{\min}} \right\rceil$$

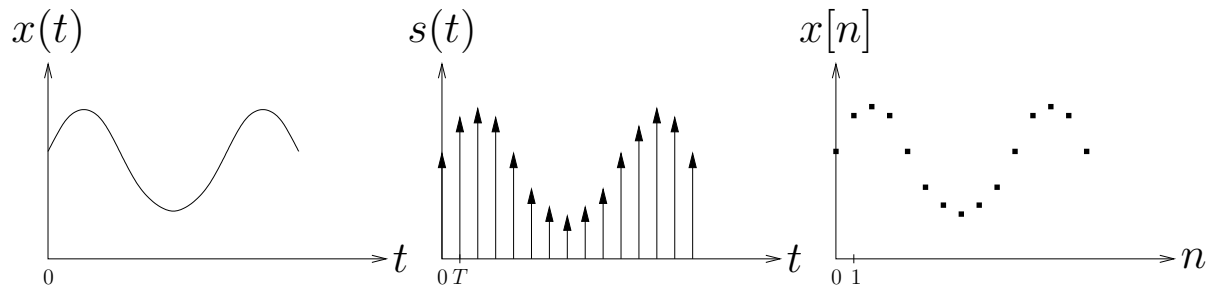
Sampling — DTFT/CTFT relationship:

To fully understand aliasing, we need to derive the relationship between the CTFT $X_c(j\Omega)$ and the DTFT $X(e^{j\omega})$.

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \\ &= \sum_{n=-\infty}^{\infty} x(nT) e^{-j\omega n} \\ &= \sum_{n=-\infty}^{\infty} \left(\int_{-\infty}^{\infty} x(t) \delta(t - nT) dt \right) e^{-j\omega n} \\ &= \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} \underbrace{x(t) \delta(t - nT) e^{-j\omega n}}_{\text{nonzero iff } n = t/T} dt \\ &= \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} x(t) \delta(t - nT) e^{-j\omega \frac{t}{T}} dt \\ &= \int_{-\infty}^{\infty} x(t) \underbrace{\sum_{n=-\infty}^{\infty} \delta(t - nT)}_{\triangleq s(t)} e^{-j\frac{\omega}{T} t} dt \\ &= S_c\left(j\frac{\omega}{T}\right) \end{aligned}$$

where we call $s(t)$ the “continuous-time sampled signal.”

Sampling — DTFT/CTFT relationship (cont):



Notice that $s(t) = x(t)p(t)$ for the T -periodic “pulse train”

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT),$$

which has the Fourier series representation

$$p(t) = \frac{1}{T} \sum_{k=-\infty}^{\infty} e^{j\frac{2\pi}{T}kt},$$

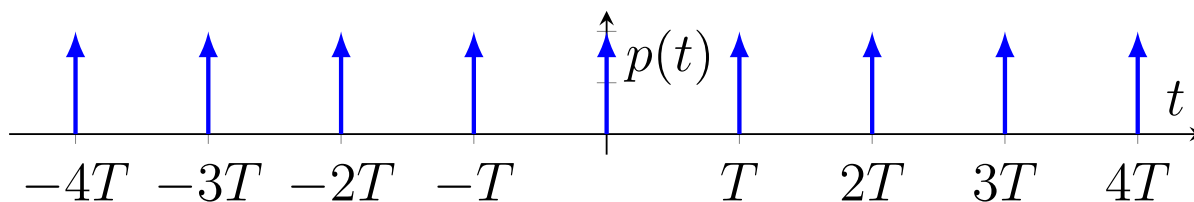
and thus

$$\begin{aligned} S_c(j\Omega) &= \mathcal{F}_{\text{CTFT}}\{s(t)\} = \mathcal{F}_{\text{CTFT}}\{x(t)p(t)\} \\ &= \int_{-\infty}^{\infty} \left(x(t) \frac{1}{T} \sum_{k=-\infty}^{\infty} e^{j\frac{2\pi}{T}kt} \right) e^{-j\Omega t} dt \\ &= \frac{1}{T} \sum_{k=-\infty}^{\infty} \int_{-\infty}^{\infty} x(t) e^{-j(\Omega - \frac{2\pi}{T}k)t} dt \\ &= \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\Omega - \frac{2\pi}{T}k)). \end{aligned}$$

Sampling — DTFT/CTFT relationship (cont):

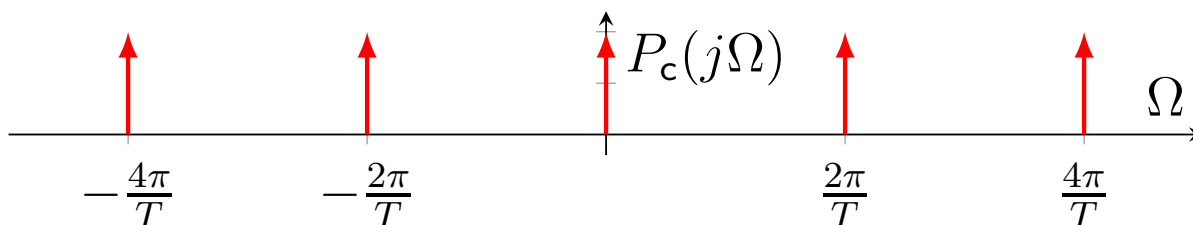
Alternative approach: Recall that the T -periodic pulse train, a.k.a. comb function, is given by

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT).$$



Since $p(t)$ is T -periodic, it is written as a Fourier series, from which its Fourier transform is derived as follows:

$$p(t) = \frac{1}{T} \sum_{k=-\infty}^{\infty} e^{j\frac{2\pi}{T}kt} \xleftrightarrow{\text{FT}} \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta\left(\Omega - \frac{2\pi k}{T}\right).$$



By the multiplication property, we obtain

$$x(t)p(t) \xleftrightarrow{\text{FT}} \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c\left(\Omega - \frac{2\pi k}{T}\right).$$

Sampling — DTFT/CTFT relationship (cont):

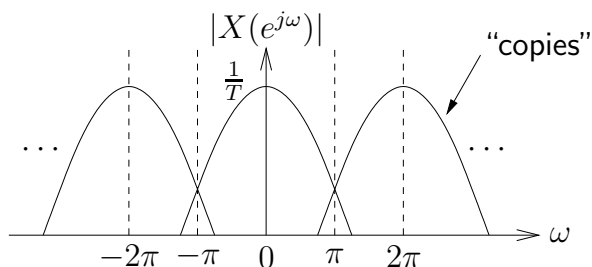
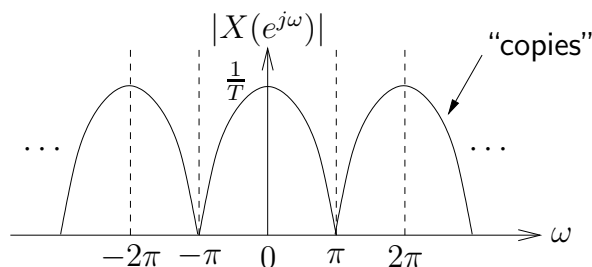
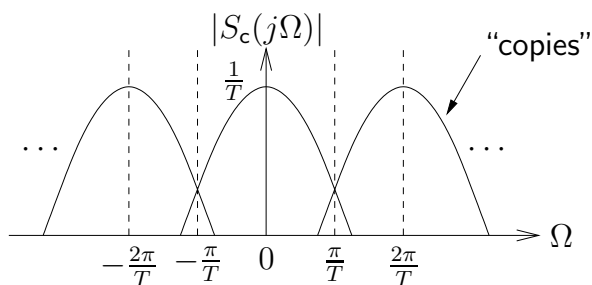
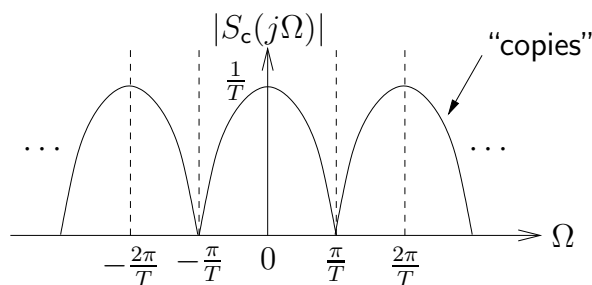
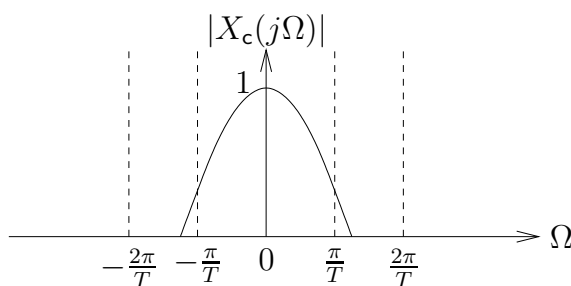
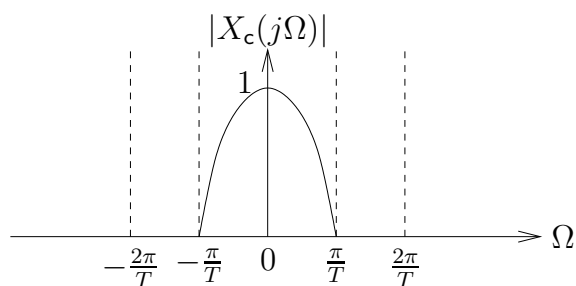
Finally, we can combine these two results into

$$X(e^{j\omega}) = S_c(j\frac{\omega}{T}) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\frac{\omega}{T} - \frac{2\pi}{T}k))$$

or, equivalently, we could substitute $\boxed{\Omega \triangleq \frac{\omega}{T}}$ to write:

$$X(e^{j\Omega T}) = S_c(j\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\Omega - \frac{2\pi}{T}k)).$$

Both can be visualized as follows:



Sampling — DTFT/CTFT relationship (cont):

In words, the equation

$$X(e^{j\omega}) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\frac{\omega-2\pi k}{T}))$$

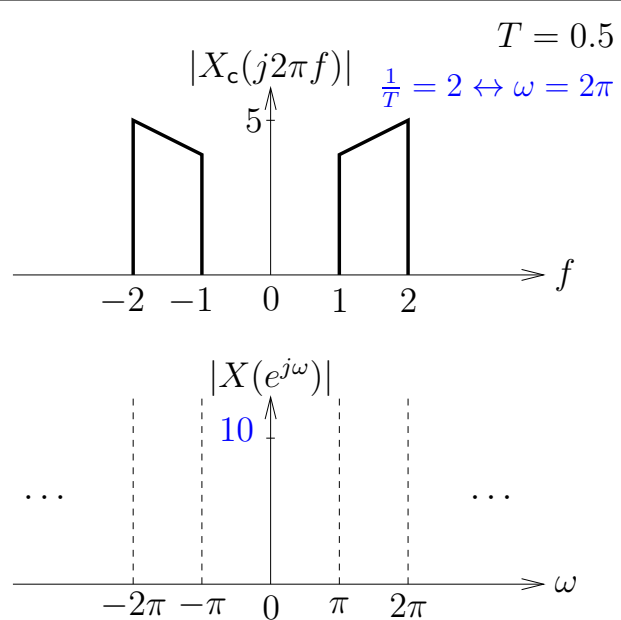
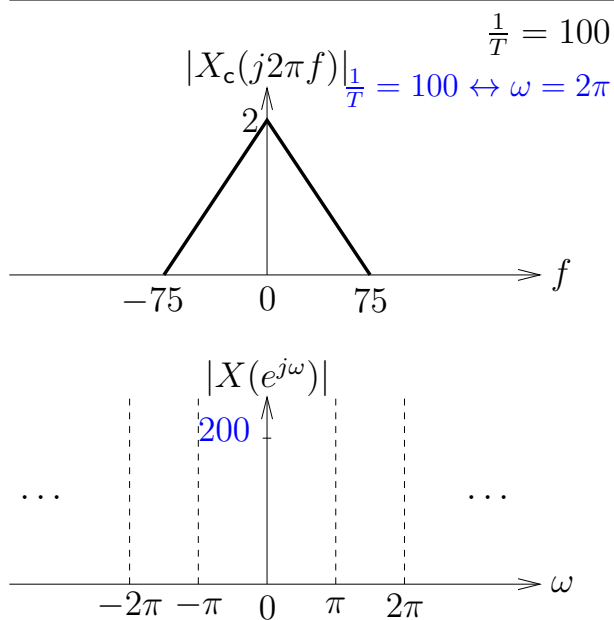
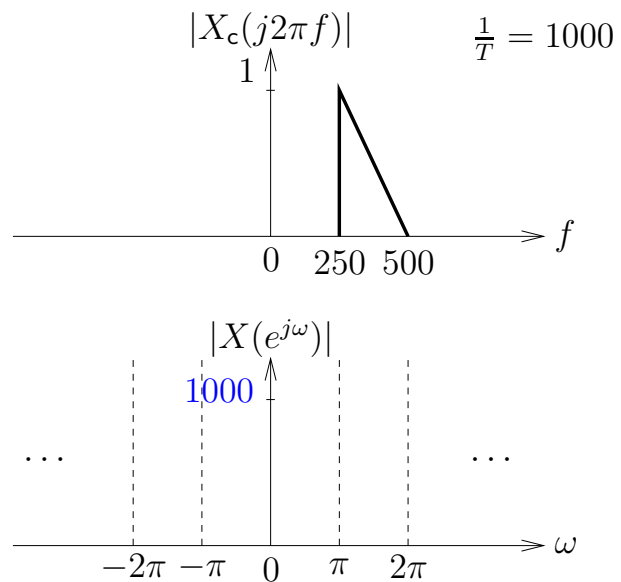
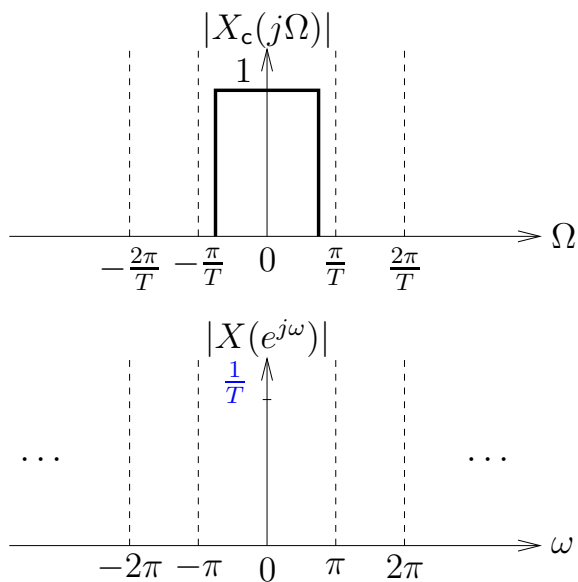
says that the DTFT spectrum $X(e^{j\omega})$ is composed of

- amplitude-scaled (by factor $\frac{1}{T}$),
- frequency-stretched (by factor T), and
- $2\pi k$ -shifted

copies of $X_c(j\Omega)$. These copies ensure 2π -periodicity.

Note: The equation above gives a complete characterization of $\frac{1}{T}$ -rate sampling, including the “aliasing” phenomenon seen earlier using sinusoidal examples.

Sampling — Examples:



Remember:

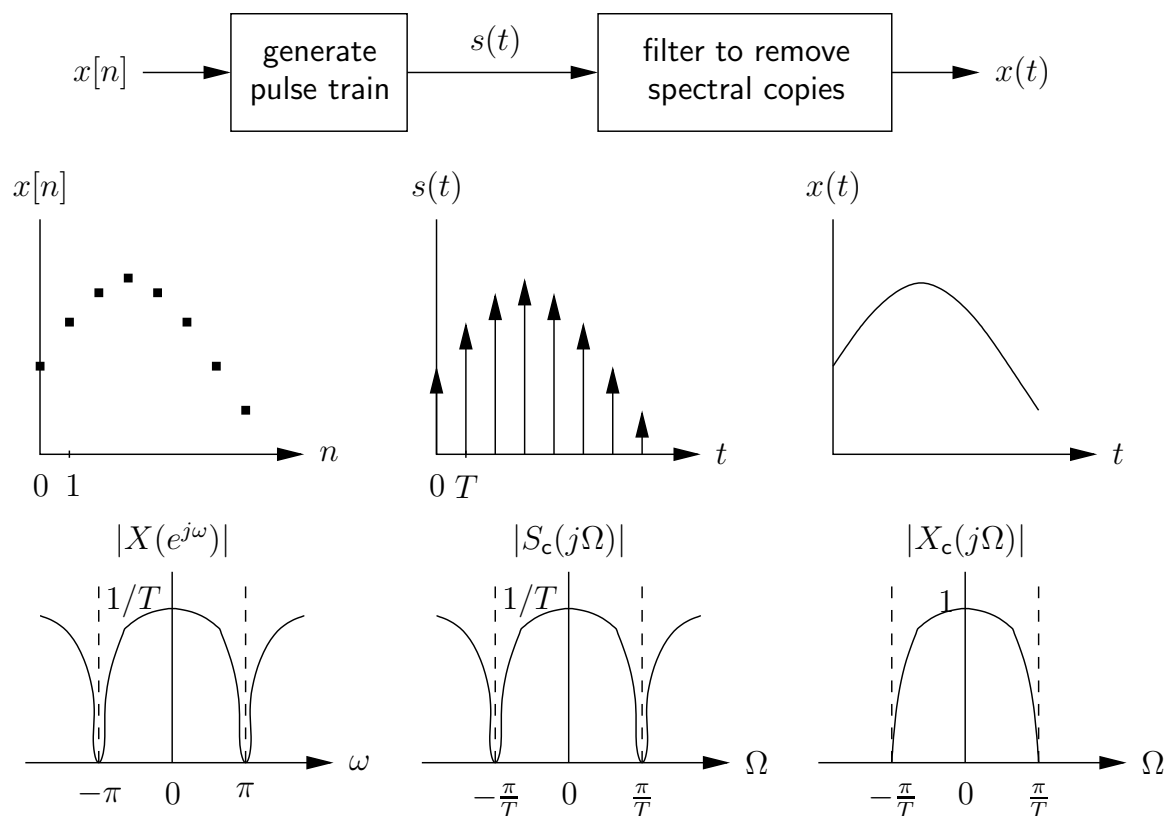
The sampling freq $\begin{cases} f = \frac{1}{T} \text{ Hz} \\ \Omega = \frac{2\pi}{T} \frac{\text{rad}}{\text{sec}} \end{cases}$ maps to $\omega = 2\pi \frac{\text{rad}}{\text{samp}}$.

Sinc Reconstruction:

- We will now detail the reconstruction of waveform $\{x(t)\}_{\forall t}$ from its samples $\{x[n]\}_{\forall n}$.
- Throughout, we'll assume that $x(t)$ is a lowpass signal (i.e., it lives in the 0^{th} Nyquist zone).
- For our CTFT/DTFT derivation, we viewed sampling as:

$$x(t) \rightarrow s(t) \rightarrow x[n].$$

We will now reverse this procedure for reconstruction.



Note: the frequency-domain plots suggest exactly the form of filtering that we need!

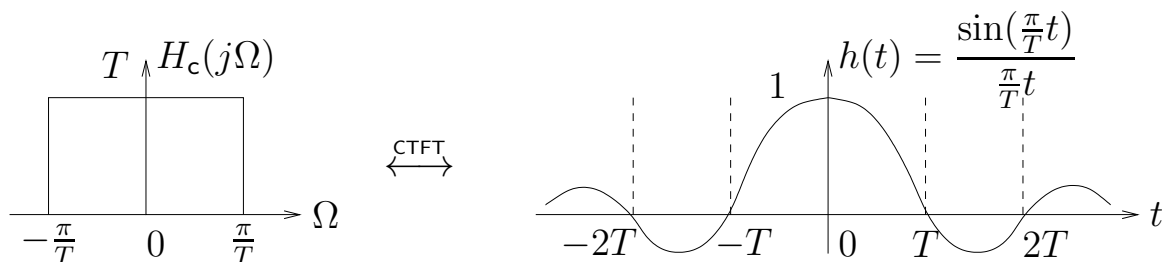
Sinc Reconstruction (cont.):

- To generate the pulse train $s(t)$, we can (in theory) do

$$s(t) = \sum_{n=-\infty}^{\infty} x[n]\delta(t - nT),$$

which places a $x[n]$ -scaled Dirac delta at all times $t = nT$.

- To obtain $x(t)$ from $s(t)$, the CTFT plots show that we need a “brickwall” lowpass filter with cutoff freq $\Omega_0 = \frac{\pi}{T}$ and gain T :



- Actually, we can combine these two operations into one:

$$\begin{aligned} x(t) &= \int_{-\infty}^{\infty} s(\tau)h(t - \tau)d\tau \\ &= \int_{-\infty}^{\infty} \sum_n x[n]\delta(\tau - nT) \frac{\sin(\frac{\pi}{T}(t - \tau))}{\frac{\pi}{T}(t - \tau)} d\tau \\ &= \sum_n x[n] \frac{\sin(\frac{\pi}{T}(t - nT))}{\frac{\pi}{T}(t - nT)} \end{aligned}$$

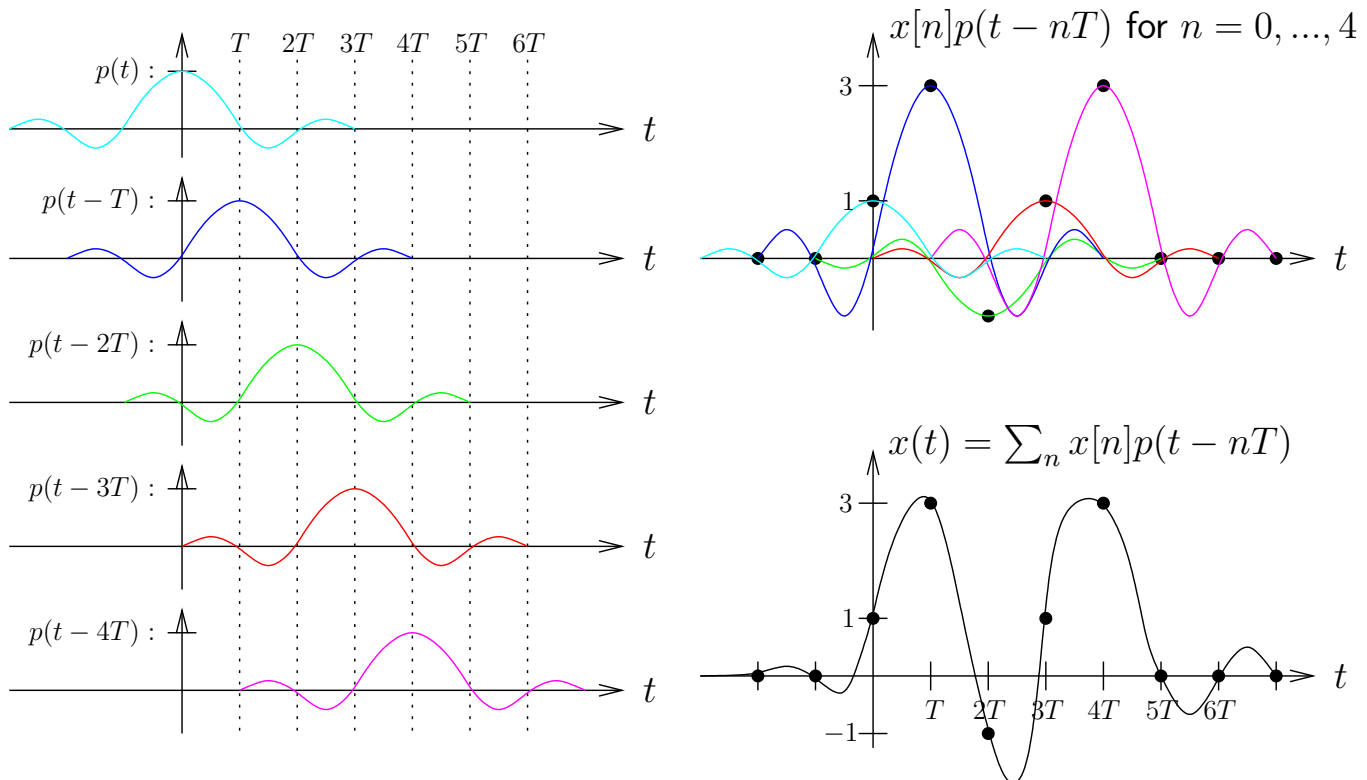
known as “sinc reconstruction.”

Sinc Reconstruction (cont.):

Sinc reconstruction:

$$x(t) = \sum_{n=-\infty}^{\infty} x[n] \frac{\sin\left(\frac{\pi}{T}(t - nT)\right)}{\frac{\pi}{T}(t - nT)}$$

interpolates the samples $\{x[n]\}$ using a sinc pulse “ $p(t)$ ” with T -spaced zero-crossings:



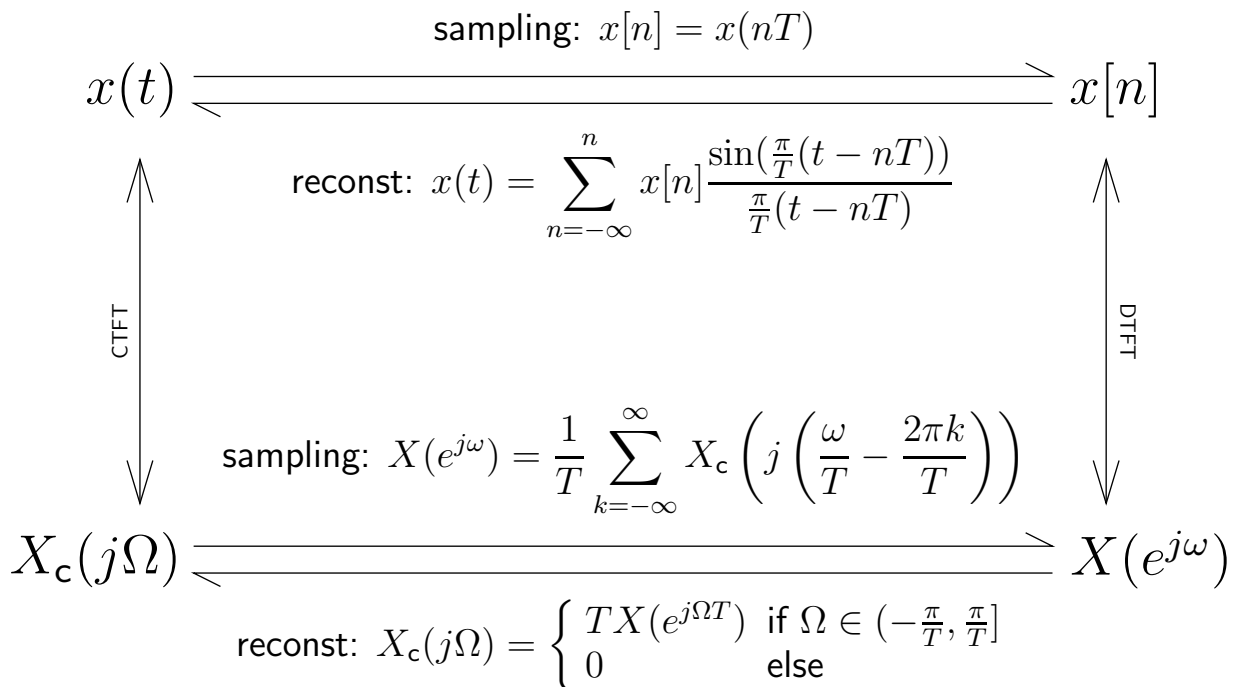
Note: the same procedure is called “Nyquist pulse shaping” in the context of digital communications.

Sampling and Reconstruction — Summary:

In the frequency domain, we saw that sinc reconstruction applies a LPF, with gain- T and cutoff $\frac{\pi}{T}$ rad/sec, to $S_c(j\Omega)$. We know $S_c(j\Omega) = X(e^{j\Omega T})$, and thus when $x(t)$ is bandlimited to $\frac{1}{2T}$ Hz, so that no aliasing occurs,

$$X_c(j\Omega) = \begin{cases} TX(e^{j\Omega T}) & \text{if } \Omega \in (-\frac{\pi}{T}, \frac{\pi}{T}] \\ 0 & \text{else.} \end{cases}$$

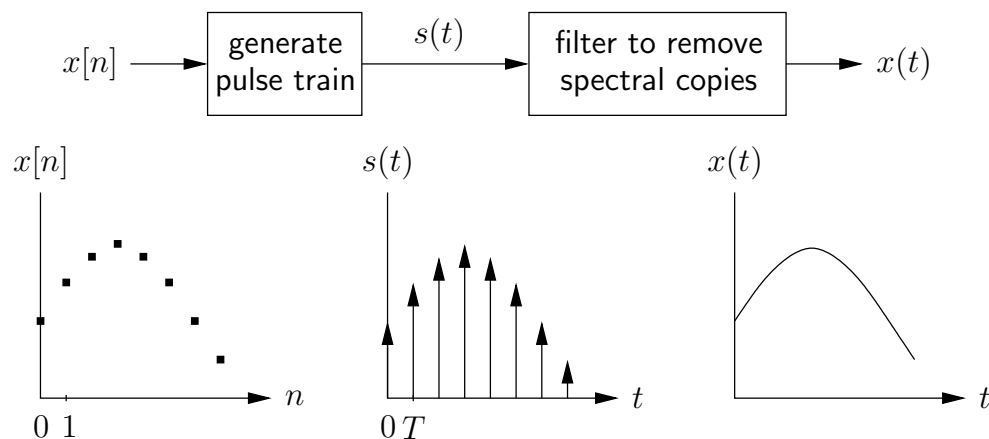
In summary, we have



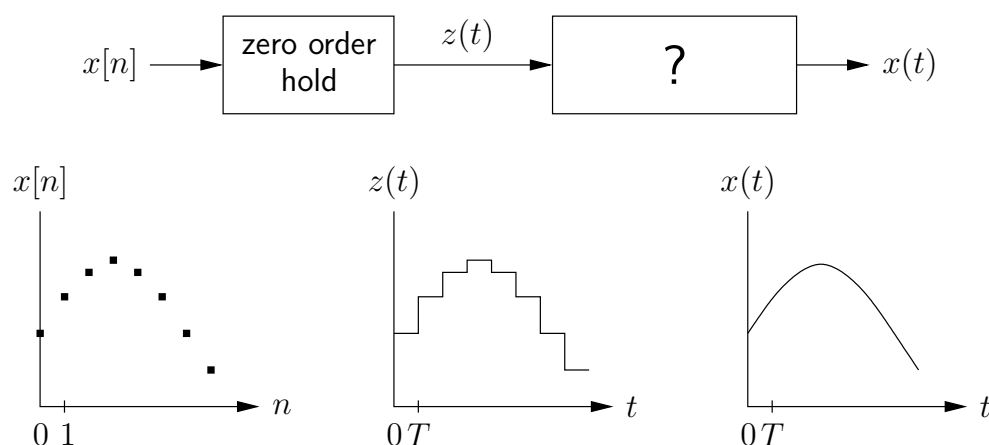
where the reconstruction equations hold *only* when $x(t)$ is bandlimited to $\frac{1}{2T}$ Hz.

ZOH Reconstruction:

- The “CT sampled signal” $s(t)$, a Dirac-delta pulse train, is impractical to generate. (It has infinite amplitude!)



- Practical Idea: Use a rectangular pulse to generate a “zero order hold” signal $z(t)$ instead of a pulse train $s(t)$.



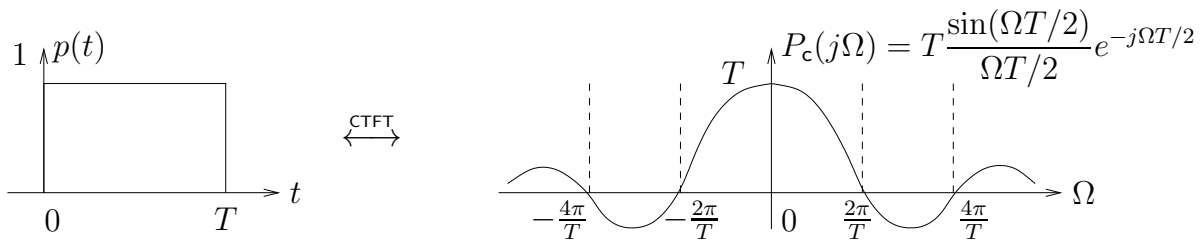
Is perfect reconstruction still possible? If so, how?

ZOH Reconstruction (cont.):

The ZOH signal $z(t)$ is related to the samples $x[n]$ via

$$z(t) = \sum_{n=-\infty}^{\infty} x[n]p(t - nT)$$

where $p(t)$ is a T -wide rectangular pulse:



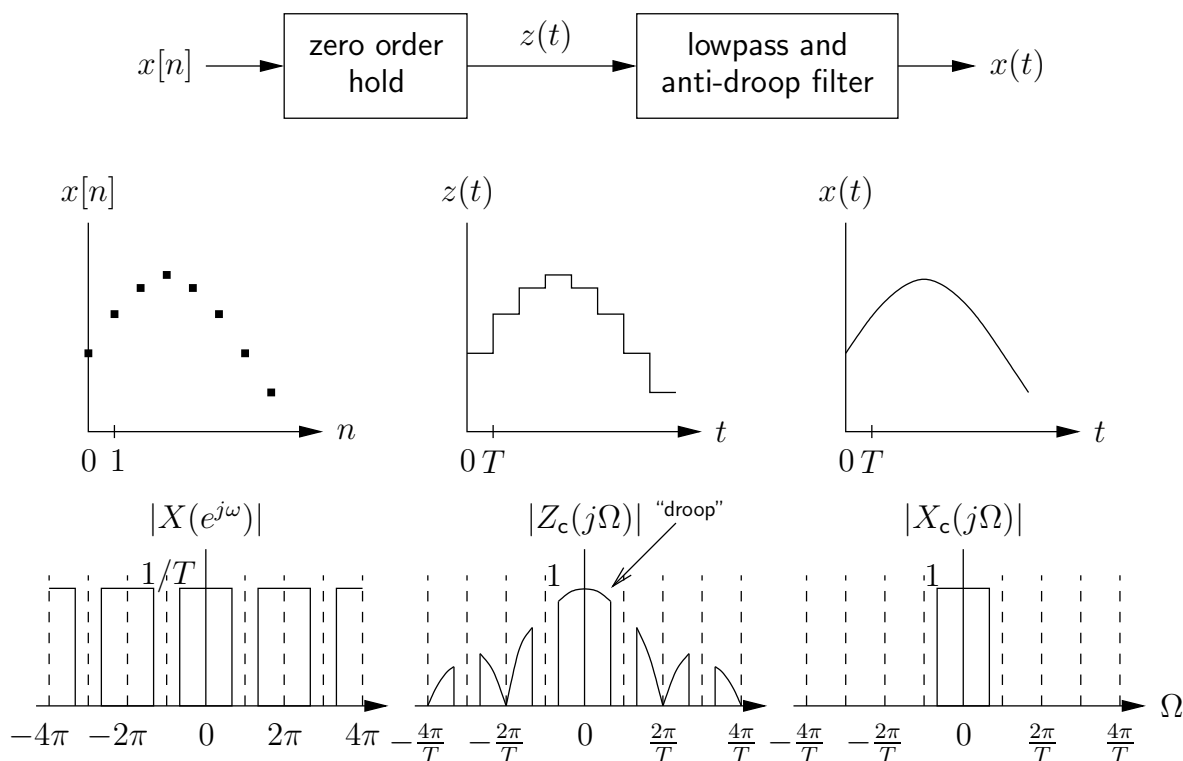
In the frequency domain, we find

$$\begin{aligned} Z_c(j\Omega) &= \int_{-\infty}^{\infty} \left(\sum_n x[n]p(t - nT) \right) e^{-j\Omega t} dt \\ &= \sum_n x[n] \int_{-\infty}^{\infty} p(t - nT) e^{-j\Omega t} dt \\ &= \sum_n x[n] e^{-j\Omega T n} P_c(j\Omega) \\ &= X(e^{j\Omega T}) \cdot T \frac{\sin(\Omega T/2)}{\Omega T/2} e^{-j\Omega T/2} \\ &= \frac{\sin(\Omega T/2)}{\Omega T/2} e^{-j\Omega T/2} \sum_{k=-\infty}^{\infty} X_c(j(\Omega - \frac{2\pi}{T}k)). \end{aligned}$$

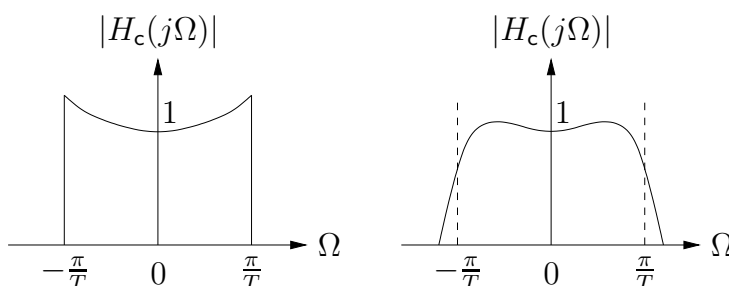
Comparing to $S_c(j\Omega)$, the effect of ZOH becomes clear!

ZOH Reconstruction (cont.):

Summary of ZOH reconstruction:



The 2nd stage of ZOH reconstruction is filtering that suppresses spectral copies and corrects for passband “droop”. Ideal (left) and practical (right) versions of this LPF look like:



where the passband gain on the left is $\left| \frac{\sin(\Omega T/2)}{\Omega T/2} \right|^{-1}$.

Rate Conversion:

- Sometimes we would like to change the effective sampling rate of a signal *after* it has been sampled.
- To do this, we combine one of the elementary operations
 - *downsampling* (throwing away samples)
 - *upsampling* (inserting zero-valued samples)

with discrete-time lowpass filtering in order to perform

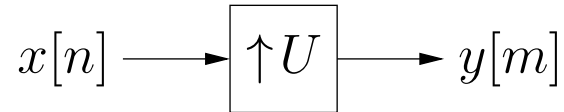
- *decimation* (decreasing effective sampling rate) and
- *interpolation* (increasing effective sampling rate),

respectively. These last two procedures change the effective sampling rate by an integer factor.

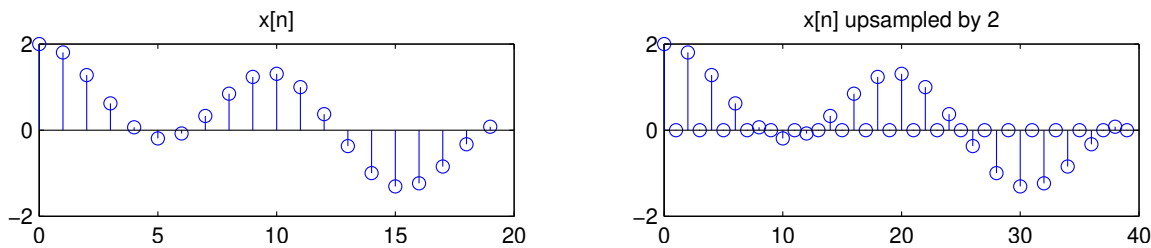
- When we want to change the effective sampling rate by a rational factor $\frac{U}{D}$, we combine interpolation and decimation into what is known as *resampling*.

Upsampling:

- We represent upsampling by the block diagram



- To upsample $x[n]$ by $U \in \mathbb{Z}^+$, we simply insert $U - 1$ zeros between each sample in $x[n]$.



- In the time domain, upsampling can be written as

$$y[m] = \begin{cases} x[\frac{m}{U}] & \text{if } \frac{m}{U} \in \mathbb{Z} \\ 0 & \text{else} \end{cases}$$

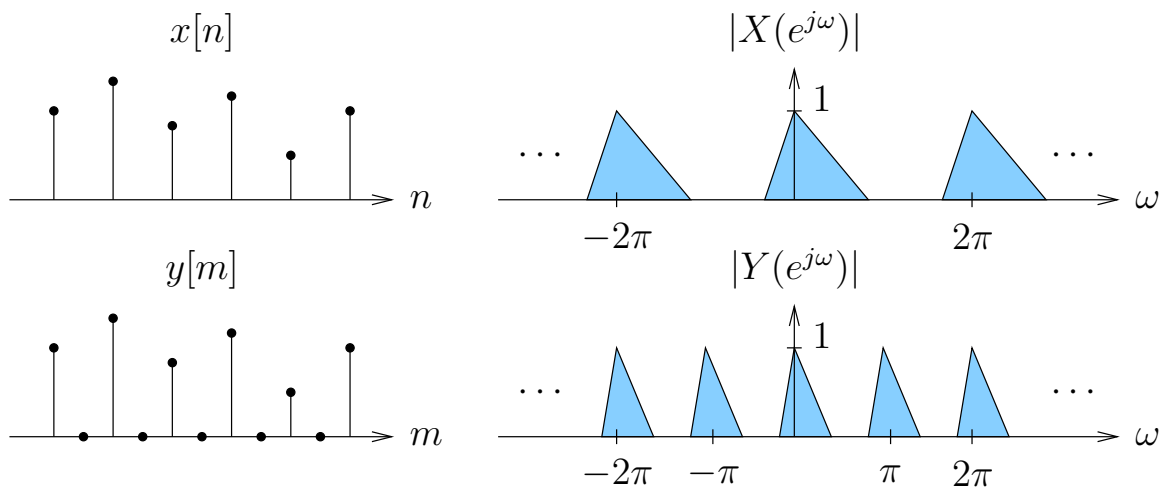
- Is upsampling
 - linear? **yes**
 - time-invariant? **no**
 - causal? **yes**
 - stable? **yes**

Upsampling (cont.):

In the frequency domain, upsampling can be written as

$$\begin{aligned} Y(e^{j\omega}) &= \sum_{m=-\infty}^{\infty} y[m] e^{-j\omega m} \\ &= \sum_{m: \frac{m}{U} \in \mathbb{Z}} x\left[\frac{m}{U}\right] e^{-j\omega m} \quad (\text{e.g., } m = \dots, 0, U, 2U, \dots) \\ &= \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega U n} \\ &= X(e^{j\omega U}) \end{aligned}$$

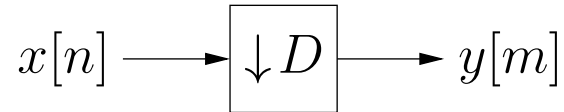
For example, when $U = 2$,



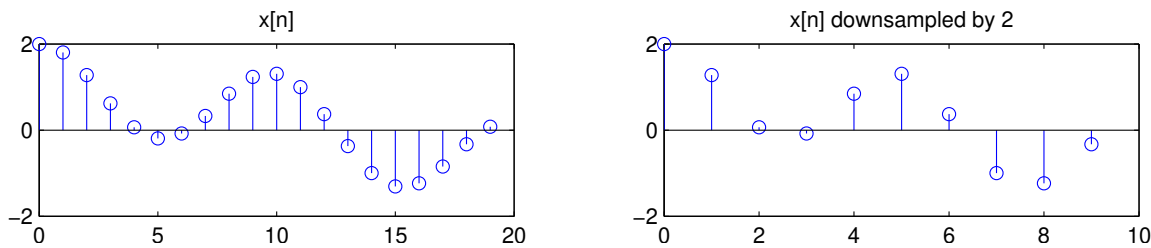
The spectrum is simply compressed by the integer factor U .

Downsampling:

- We represent downsampling by the block diagram



- To downsample $x[n]$ by $D \in \mathbb{Z}^+$, we keep every D^{th} sample of $x[n]$ and discard the others.



- In the time domain, downsampling can be written as

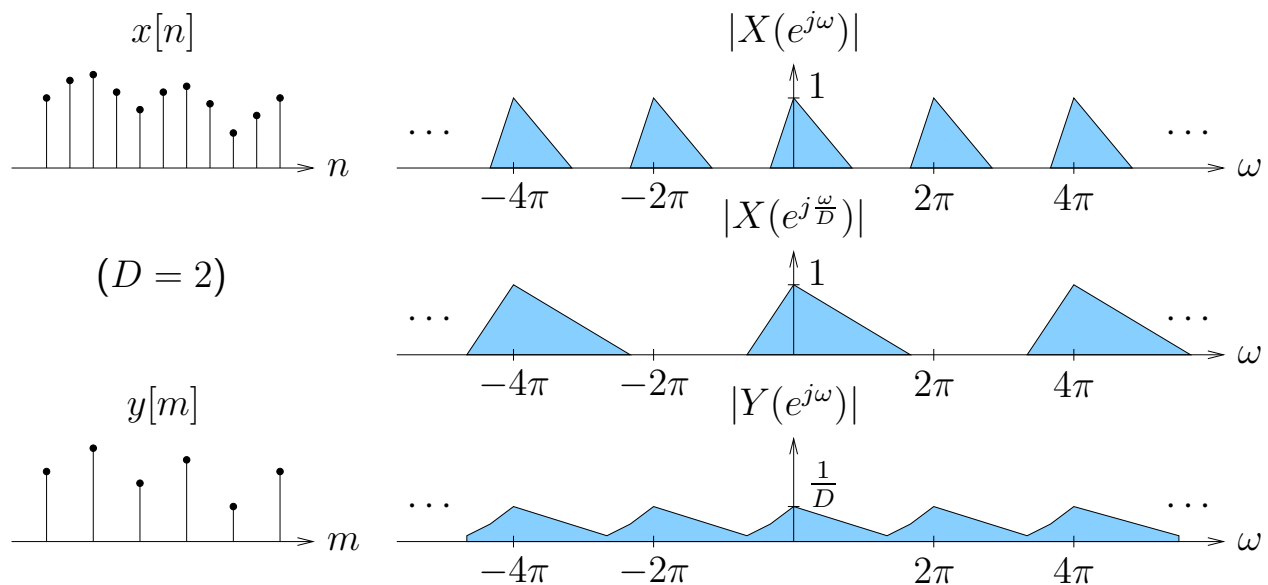
$$y[m] = x[Dm]$$

- Is downsampling
 - linear? **yes**
 - time-invariant? **no**
 - causal? **yes**
 - stable? **yes**

Downsampling (cont.):

In the frequency domain, downsampling can be written as

$$\begin{aligned}
 Y(e^{j\omega}) &= \sum_{m=-\infty}^{\infty} x[mD]e^{-j\omega m} = \sum_{n=-\infty}^{\infty} x[n] \underbrace{\delta[\langle n \rangle_D]}_{= 1 \text{ iff } n \text{ is a mult of } D} e^{-j\omega \frac{n}{D}} \\
 &= \sum_{n=-\infty}^{\infty} x[n] \left(\frac{1}{D} \sum_{k=0}^{D-1} e^{j\frac{2\pi}{D}kn} \right) e^{-j\frac{\omega}{D}n} \quad (\text{next page!}) \\
 &= \frac{1}{D} \sum_{k=0}^{D-1} \sum_{n=-\infty}^{\infty} x[n] e^{-j(\frac{\omega-2\pi k}{D})n} \\
 &= \frac{1}{D} \sum_{k=0}^{D-1} X(e^{j\frac{\omega-2\pi k}{D}})
 \end{aligned}$$



Spectrum is expanded by D , scaled by $\frac{1}{D}$, and made periodic.

Useful Facts:

Why is $\delta[\langle n \rangle_D] = 1$ iff n is an integer mult of D ?

If $D = 3$,

n	-3	-2	-1	0	1	2	3	4	5
$\langle n \rangle_D$	0	1	2	0	1	2	0	1	2
$\delta[\langle n \rangle_D]$	1	0	0	1	0	0	1	0	0

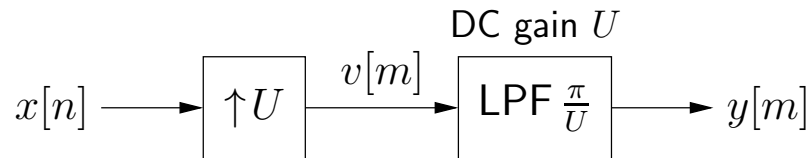
Why is $\frac{1}{D} \sum_{k=0}^{D-1} e^{j\frac{2\pi}{D}kn} = \delta[\langle n \rangle_D]$? (Discrete Poisson sum.)

Draw pictures for $D = 3$ and the following four cases:

$$n = 0, 1, 2, 3.$$

Interpolation:

- Interpolation goal: increase effective sampling rate from $\frac{1}{T}$ Hz to $\frac{U}{T}$ Hz. We can accomplish this in two steps:



- To understand the choice of LPF, recall that

$$X(e^{j\omega}) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c\left(j\left(\frac{\omega - 2\pi k}{T}\right)\right)$$

$$Y(e^{j\omega}) = \frac{U}{T} \sum_{k=-\infty}^{\infty} X_c\left(j\left(\frac{\omega - 2\pi k}{T/U}\right)\right),$$

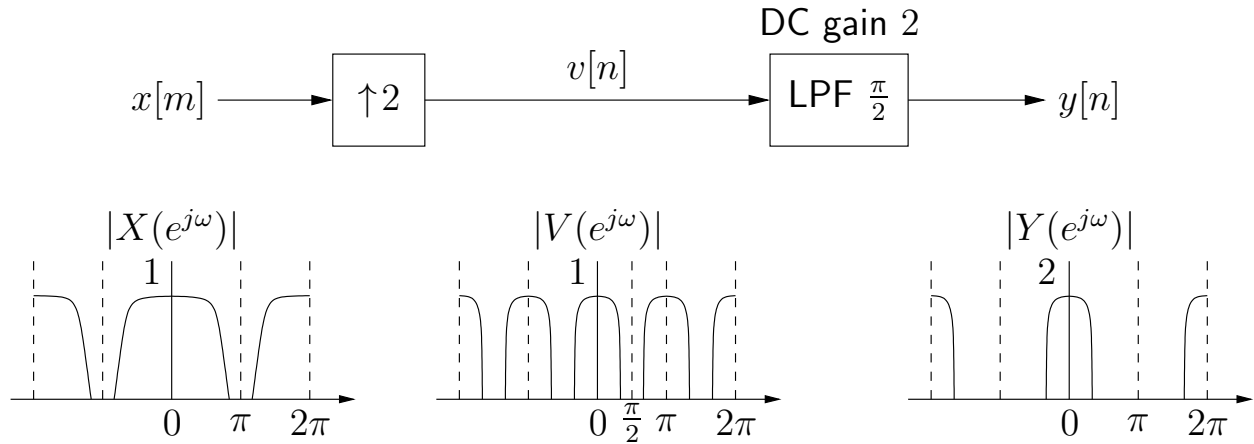
and notice that

$$\begin{aligned} V(e^{j\omega}) &= X(e^{j\omega U}) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c\left(j\left(\frac{\omega U - 2\pi k}{T}\right)\right) \\ &= \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c\left(j\left(\frac{\omega - \frac{2\pi}{U}k}{T/U}\right)\right). \end{aligned}$$

We see that $V(e^{j\omega})$ has spectral copies at $\omega \in \{\frac{2\pi}{U}k\}_{\forall k}$ while $Y(e^{j\omega})$ has them at $\omega \in \{2\pi k\}_{\forall k}$. Thus, digital LPFing is needed to suppress the unwanted spectral copies (and scale by U).

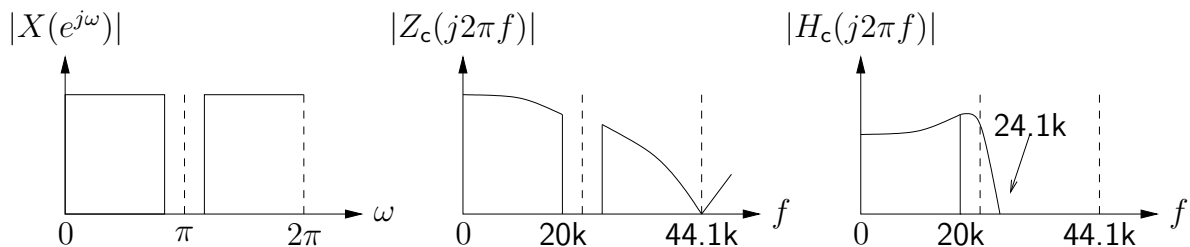
Interpolation (cont.):

- Interpolation example for $U = 2$:

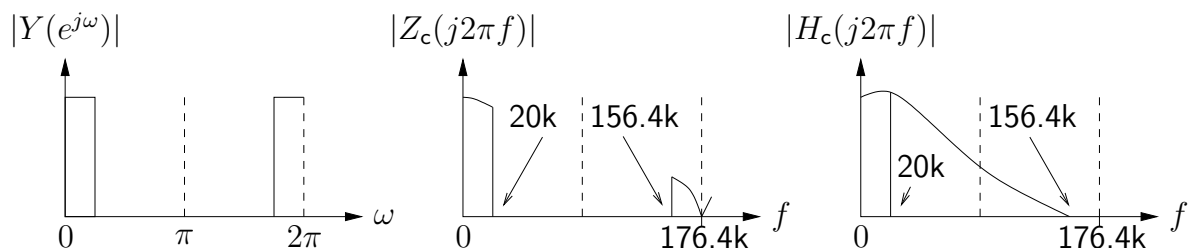


LPFing removes the copies centered at $\omega \in \{\pi k\}_{\forall \text{ odd } k}$.

- Application — CD-Audio D/A conversion:
 - ZOH reconstruction without oversampling:



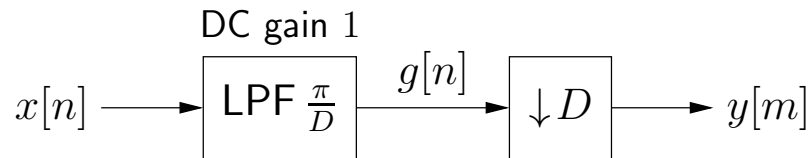
- ZOH reconstruction after $4\times$ oversampling:



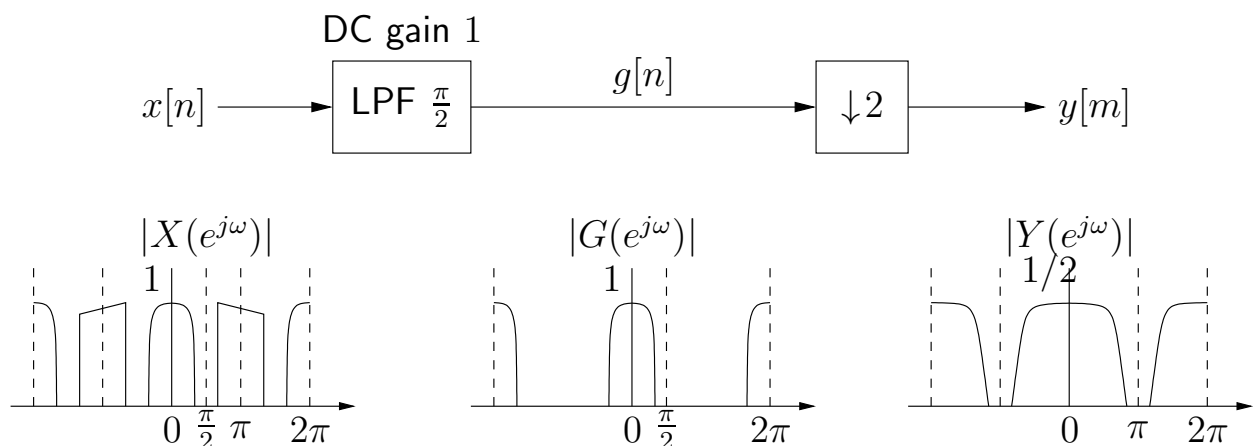
Analog filter $H_c(j\Omega)$ becomes much cheaper!

Decimation:

- Decimation goal: decrease effective sampling rate from $\frac{1}{T}$ Hz to $\frac{1}{DT}$ Hz. To prevent aliasing, we LPF first:

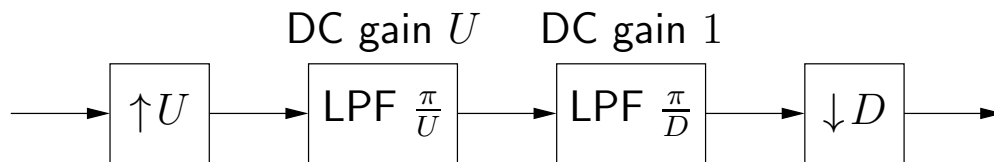


- To understand the LPFing, say that $x[n] = x(nT)$ with $x(t) = g(t) + b(t)$,
 $g(t)$: lowpass and bandlimited to $\frac{1}{2DT}$ Hz
 $b(t)$: bandpass between $\frac{1}{2DT}$ and $\frac{1}{2T}$ Hz
 - For $\frac{1}{DT}$ -rate sampling, $g(t)$ satisfies Nyquist thm, but $b(t)$ does not, so we would have suppressed $b(t)$.
 - Here, $x[n] = g[n] + b[n]$, where $b[n]$ will alias in downsampling but $g[n]$ will not. Thus, suppress $b[n]$.
- Decimation example for $D = 2$:

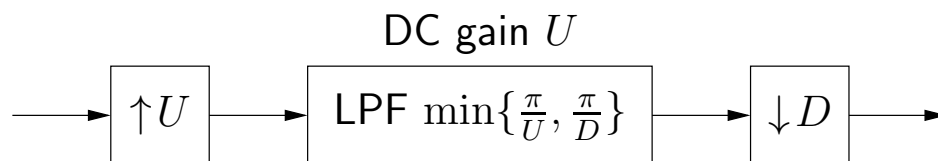


Resampling by a rational factor:

- To resample by the rational factor $\frac{U}{D}$, we first interpolate by U and then decimate by D



which can be simplified to



- Example — CD to DAT rate conversion:
 - CD sampling rate = 44.1 kHz
 - DAT sampling rate = 48 kHz.
 - Resampling ratio is $\frac{U}{D} = \frac{480}{441} = \frac{160}{147}$.
- Note:
 - The LPF filter will operate on many zero-valued input samples, and many of its outputs will be thrown away!
 - Efficient “polyphase” filter structures can avoid these redundant computations.